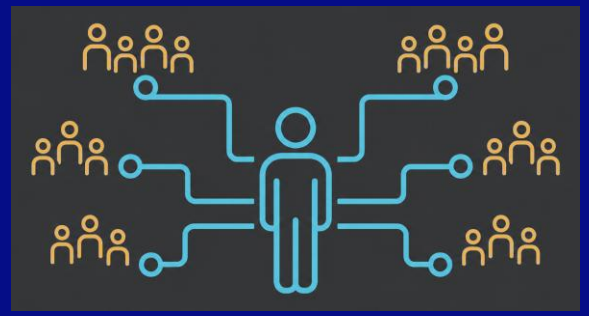


DHUM101

Socially Responsible Computing

University of Otago
ŌTĀKOU WHAKAIHU WAKA
Te Kura Rorohiko
School of Computing
Te Reo Inarahi me te Mātai Wetereo
English and Linguistics

Lecture 6
Consequences and Social Impact

- Hunter Hatfield
Semester 2, 2026



Today

- The Consequences of Our Actions
- Social Impacts and Marginalisation
- How Can We Design to Reduce Social Harm?



The Consequences of Our Actions



How Do We Know What Is Right Or Wrong?

There is no right or wrong. There is only power. OR It's all opinion.

Society decides. Morality is a social convention.

Follow the model of wise people.

Rights and duties of actions. (Kant's system is a version of this.)

Determined by the consequences of actions. Consequentialism



Consequentialism

- Morality is based upon the consequences of your actions
- Operates like an optimisation algorithm
- What to optimise? Pleasure, happiness, fulfilment, etc.
- What is the optimisation function? Summation, something else?



Consequentialism

What is the good thing you want the most of?

Who should get the good thing?



Consequentialism

- Happiness or pleasure
- Higher-order pleasures
- Peace
- Social harmony
- Material wealth

Who should get the good thing?



Consequentialism

What is the good thing you want the most of?

- All people equally
- People in your group
- People close to you
- People who are more valuable
- People who contribute a lot



Example Cases

- Maximise: Happiness
- Who: All People Equally



Example Cases

- Maximise: Happiness
- Who: All People Equally

- Gifts to ten people is better than gifts to five people.



Example Cases

- Maximise: Happiness
- Who: All People Equally

- Feeding ten people is better than feeding five people.



Example Cases

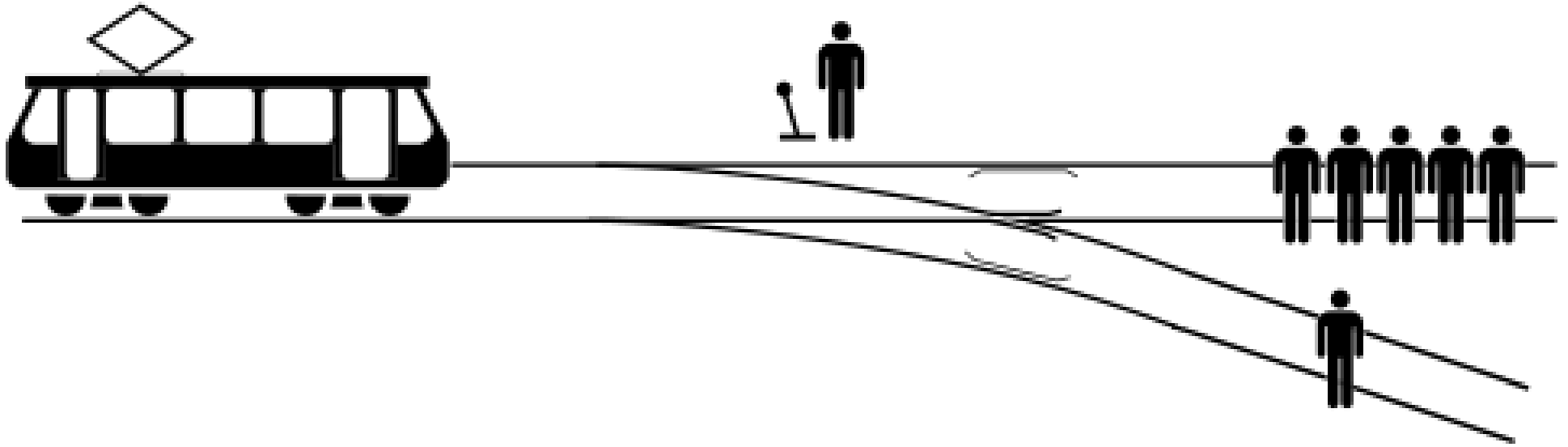
- Maximise: Happiness
- Who: All People Equally

- Feeding ten people is better than feeding five people.
- Feeding ten strangers is better than feeding five members of your family.
- Feeding ten murderers is better than feeding five infants



Example Cases

- The Trolley Problem





By Original: McGeddon Vector: Zapyon - This SVG diagram includes elements from this icon:, CC BY-SA 4.0, https://commons.wikimedia.org/w/index.php?curid=67107784

Example Cases

- Helpful for designing self-driving cars?
- MIT Team: Moral Machine



An Ethical Dilemma

You are a sheriff in an unfortunate town.

- A mob thinks that a man – Randy – has committed a horrible crime.
- They want 'justice'.
- You know that Randy didn't do it but you have no way to convince the mob.
- What do you do?
  - If you turn over Randy, one innocent person will be hurt.
  - If you do NOT turn over Randy, many innocent people will be hurt.
- This sort of thing does happen
- What is the right thing to do? – Consequentialism says it is best to save the more innocent people, even if you have to frame Randy.



Conclusions

- Consequences matter.
- Dilemma arises when intentions and consequences don't match.
  - You act intending to make things better
  - The result is things are worse
- A very common place where these break down is with marginalisation.
  - A system exists that excludes or de-prioritises some groups
  - When a person takes action in such a system, they may know less how it will affect marginalised groups.



What is the best thing to do?

Work towards a world where these difficult choices are minimised.

It is the impossible situations that are the problem.



Social Impacts and Marginalisation



Marginalisation

Marginalization. A person or group on the edges (margins) of society such that they have less power, access, or agency and therefore often do not gain the full, common benefits of the society.

Common (definitely not comprehensive dimensions):

- Age
- Ethnicity
- Gender and Gender Identity
- Sexuality / Orientation
- Disability
- Religion
- Class



These dimensions will vary, expand, and contract depending on the context.

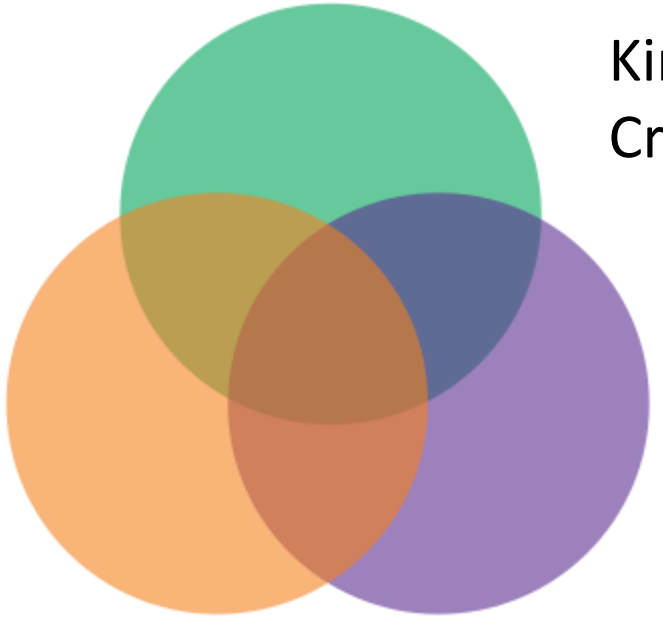
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Intersectionality



Kimberlé Crenshaw

These dimensions will vary, expand, and contract depending on the context.

Marginalisation

Biased Data. A bias in data that (a) does not accurately represent what it is intended to, and/or (b) results in outcomes other than intended.

Dark Rabbit Holes. Moving towards a negative, often unwanted experience with a technology.

Malicious Use / Abusability. Systems can be used maliciously, and malicious use often is targeted towards or more greatly affects marginalised groups.



Biased Data

How to implement biased data in three easy steps. An example

Create an algorithm to use historical data to predict the future. Example: Pre-process job applications for human review.

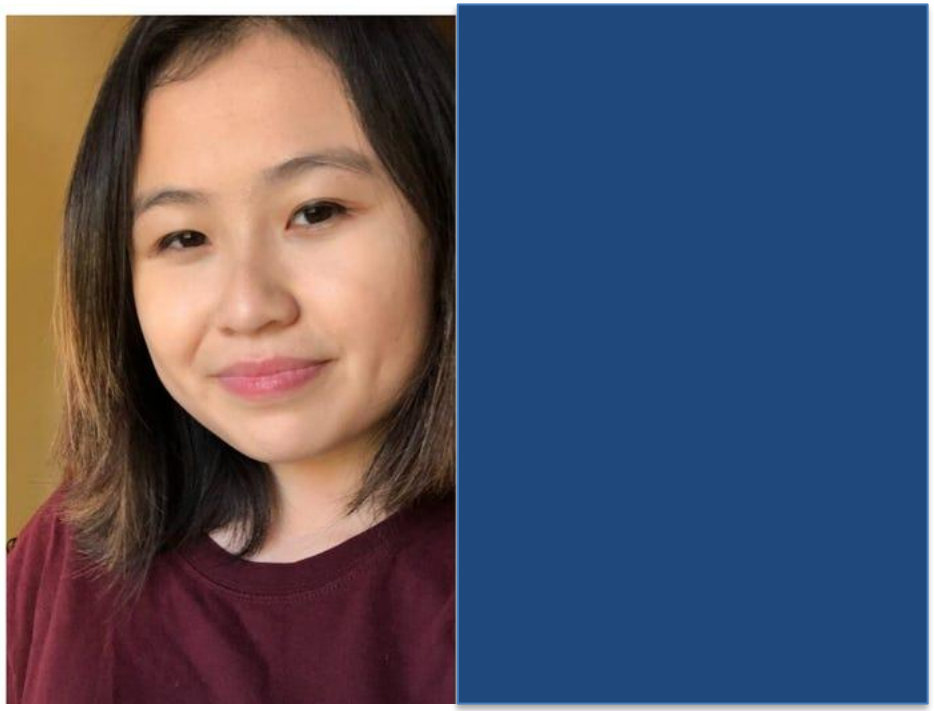
Load historical data. Example: Historically, people with English sounding names had these jobs.

Result. The algorithm only sends through applicants with English sounding names for further consideration.



Biased Data

MIT graduate uploaded a selfie and asked Playground AI to make it look professional for LinkedIn.



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https://www.businessinsider.com/student-uses-playrgound-ai-for-professional-headshot-turned-white-2023-8

Biased Data

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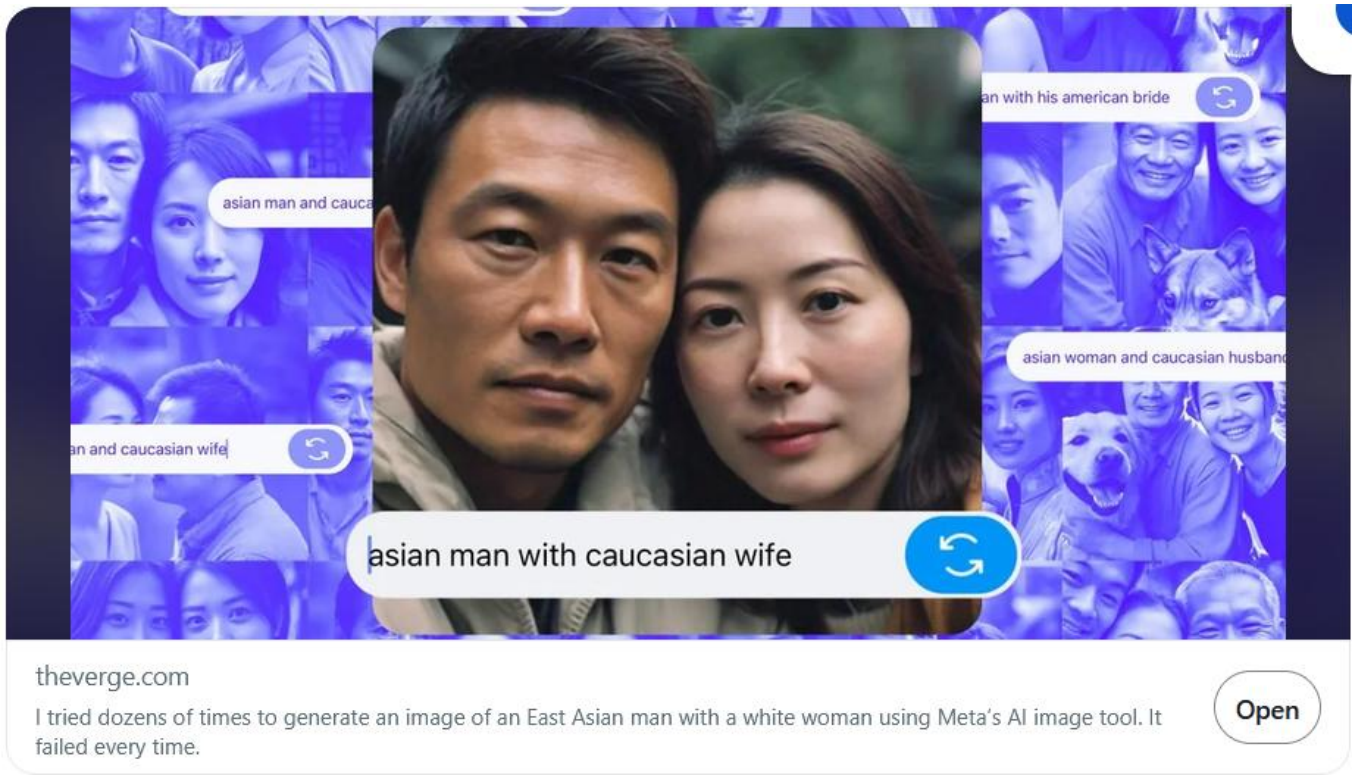
https://www.businessinsider.com/student-uses-playrgound-ai-for-professional-headshot-turned-white-2023-8

Biased Data

Ask Midjourney to create an image of a romantic couple.

Struggled with some ethnicity pairings

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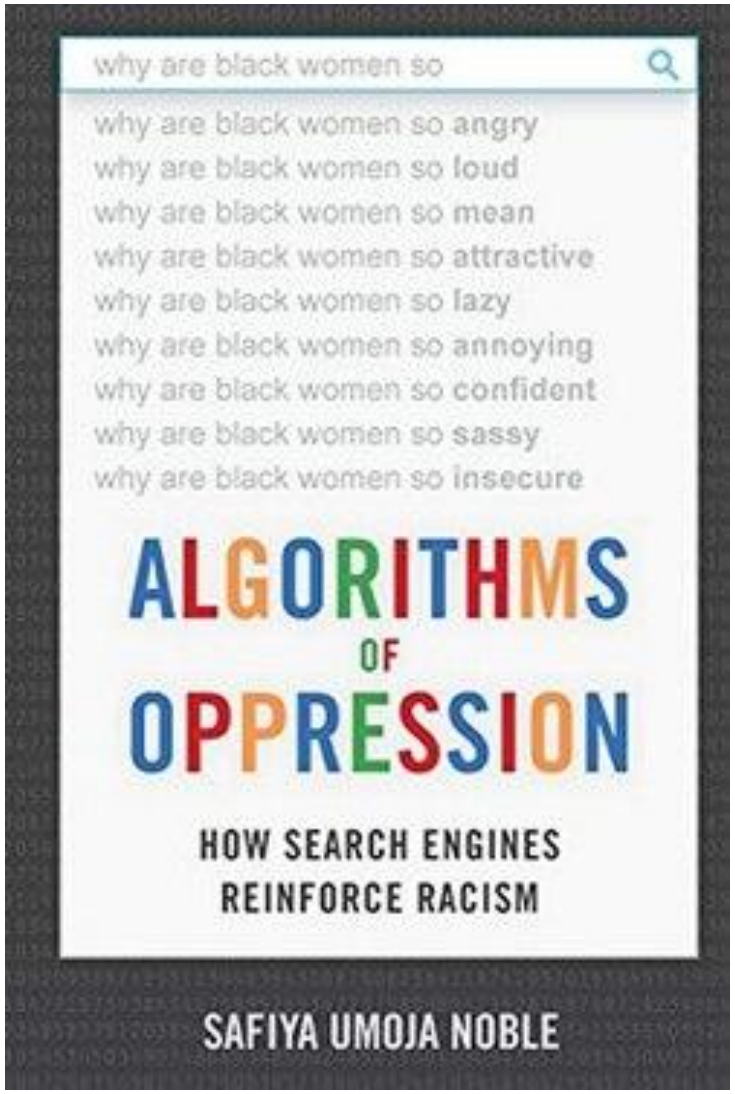


https://www.theverge.com/2024/4/10/24122072/ai-generated-asian-man-white-woman-couple-gemini-dalle-midjourney-tests

Biased Data

Biased data in an algorithm can reinforce racist discourse

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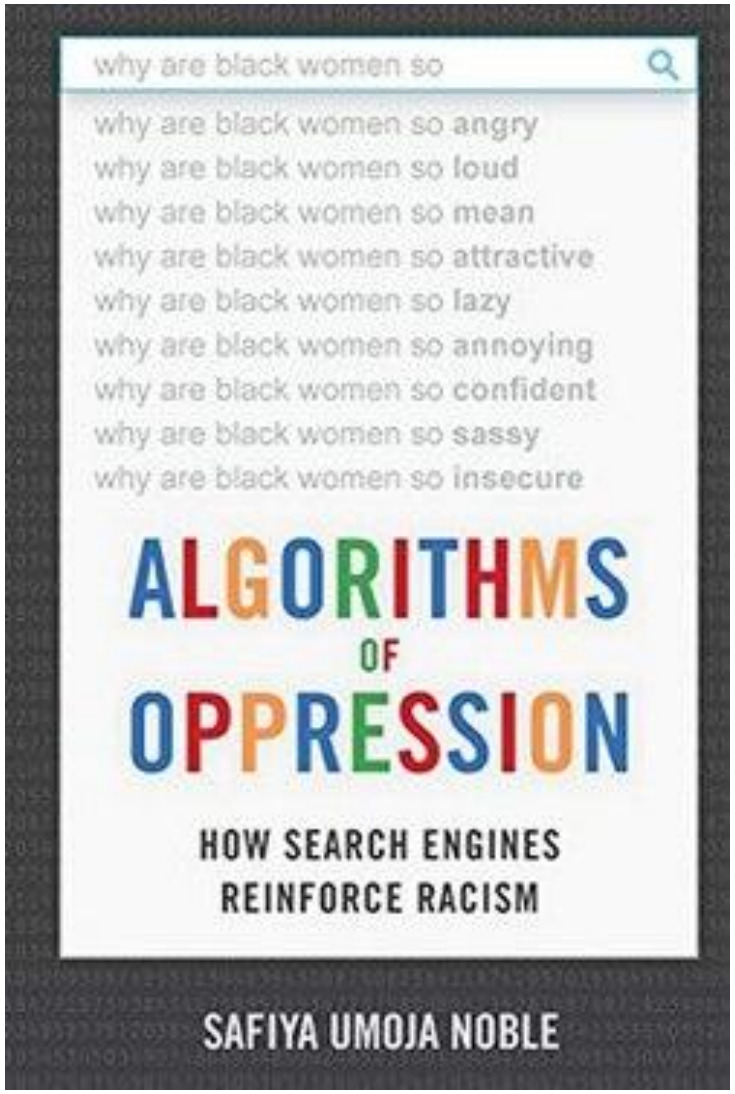
https://en.wikipedia.org/wiki/Algorithms_of_Oppression

Biased Data

Biased data in an algorithm can reinforce racist discourse

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Company Reply: This is not our intention
Advocate Reply: But this is the consequence.



https://en.wikipedia.org/wiki/Algorithms_of_Oppression

Biased Data

Meaningful Impacts.

- Tools may simply fail at higher rates for some groups.
- Biased facial recognition data can result in more false positives for criminality for some ethnicities than others.
- Algorithms promote some groups' employment applications over others for non-job-related reasons.
- Over reliance on an AI that has biased data leads to biased individual actions.
- LLMs perpetuated race-based clinical understandings in health situations.



Should we treat all data equally?

Rabbit Holes. When an algorithm takes a user to a particular subsection or experience of the data/community.



Should we treat all data equally?

Dark Rabbit Holes. When an algorithm takes a user to a particular subsection of the data/community.

Subsections may include

- Misogyny
- Homophobia
- Transphobia
- Racism
- Ableism
- Classism
- Ageism
- Violence



Should we treat all data equally?

Dark Rabbit Holes. When an algorithm takes a user to a particular subsection of the data/community.

Subsections may include

- Misogyny
- Homophobia
- Transphobia
- Racism
- Ableism
- Classism
- Ageism
- Violence

A rabbit hole does not need to be “dark” for it to have negative consequences. It might simply be so closed off that large parts of the world become unknown.



Malicious Use / Abusability

Malicious Use / Abusability. Systems can be used maliciously, and malicious use often is targeted towards or more greatly affects marginalised groups. Abusability is the ease with which a system can be used maliciously.

Can affect all. But often hurts some groups more than others.

Example: Coercive Control through Internet of Things. May allow frequent monitoring and repeated sampling of activities. Can affect anyone, but most often affects people where there are historical patterns of control and/or those in precarious situations

- Young and Older Women
- Women with Disabilities
- Migrant and Refugee Women
- LGBTIQ+ people

https://www.esafety.gov.au/industry/tech-trends-and-challenges/convergence-series/internet-of-things-iot-and-coercive-control#groups-most-at-risk



Malicious Use / Abusability

Public cameras. Police (illegally) looked at data regarding partners.

Video sharing. Can be used for revenge porn or deepfakes of intimate acts, affecting some more than others.

Comment posting. May be used to target

- LGBTIQ+ people
- Children
- People with differing physical appearance

Phishing attacks and social engineering: May target older people more.

Celebrity deepfakes. May target children and younger people more.



A Lot to Learn From

A lot of social harms comes from ignorance and exclusion.

Which means that inclusion reduces the chance of harm!

- Recall MIT student using AI for Linked-In profile
- Dr. Joy Buolamwini (Algorithmic Justice League) was attempting to do a postgrad project on facial analysis software – the software would not recognise her darker skin color but did recognise lighter skin classmates.



A Lot to Learn From

Marginalised groups often have a lot of experience negotiating adverse consequences

Transgender identity management across social media platforms

Justin Buss, Hayden Le, and Oliver L Haimson View all authors and affiliations

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Abstract

Transgender people use social media for identity work, which takes place over time and across platforms. In this study, we interviewed 20 transgender social media users to examine transgender identity management across the social media ecosystem. We found that transgender social media users curate their social media experiences to fit their needs through creating accounts on different platforms, maintaining multiple accounts on individual platforms, and making active decisions about content they post, networks they are connected to, and content they interact with. In this way, transgender people's social media curation is not limited to their own identity presentations, but also involves curating the content they see from others and whom they include in their networks. Together, these two types of online curation enable transgender social media users to craft social media worlds that meet their social and self-presentational needs.

How Can We Design to Reduce Social Harm?



How Can We Mitigate?

Stages of Development.

- Planning
- Building
- Deploying
- Monitoring

Consider both benefits and harms at all stages.



How Can We Mitigate?

Planning

- Who will be the users of the system?
  - Direct
  - Indirect
- What data will be used and where does it come from?
- Could the data be biased?
- Can the data be processed neutrally? Should it be?
- Could the system be used maliciously?



How Can We Mitigate?

Planning – Social Media

- Who will be the users of the system?
  - Direct
    - People who will post, people who will read, and 'people' who will take
    - Technological differences: geography, society, age, more.
  - Indirect
    - People not on social media are affected by social media
- What data will be used and where does it come from?
  - Posts on the system, which could be… anything? Media? Source?
- Could the data be biased?
  - Uncontrolled data suggests creating tools for users to manage it
- Can the data be processed neutrally? Should it be?
  - Should we treat all posted data with the same algorithms or adjust?
- Could the system be used maliciously?
  - Consider previous harms in media and build cases for how that would be done with new one.
  - Consider historical patterns to groups



How Can We Mitigate?

Think Broadly About Your Team

- Can the development team include a broad sample of people who will be most affected by the system?
- If not,
  - Can you consult with experts – researchers and those living it?
  - Can you read research on the area?
  - Can you set up testing and monitoring from experts?



Conclusions



Conclusions

Technology affects people for good or ill. That's why it's exciting!

Intentions are not enough; need to consider results.

Technology does not always affect all people in the same way. Every situation is unique, but history gives you a place to start.

Need a systematic way to explore likely results. Then you can build in tools to get closer to your intention.

